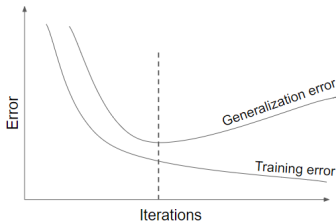


# Deep Learning

## Early Stopping

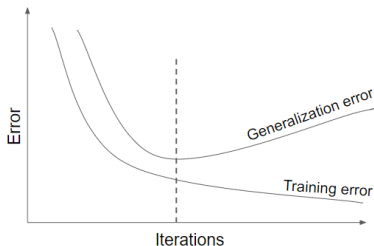


### Learning goals

- Know how early stopping works
- Understand how early stopping acts as a regularizer
- Know early stopping imitates  $L2$  regularization in some cases

# EARLY STOPPING

- When training with an iterative optimizer such as SGD, it is commonly the case that, after a certain number of iterations, generalization error begins to increase even though training error continues to decrease.
- **Early stopping** refers to stopping the algorithm early before the generalization error increases.



**Figure:** After a certain number of iterations, the algorithm begins to overfit.

# EARLY STOPPING / 2

How early stopping works:

- ➊ Split training data  $\mathcal{D}_{\text{train}}$  into  $\mathcal{D}_{\text{subtrain}}$  and  $\mathcal{D}_{\text{val}}$  (e.g. with a ratio of 2:1).
- ➋ Train on  $\mathcal{D}_{\text{subtrain}}$  and evaluate model using the validation set  $\mathcal{D}_{\text{val}}$ .
- ➌ Stop training when validation error stops decreasing (after a range of “patience” steps).
- ➍ Use parameters of the previous step for the actual model.

More sophisticated forms also apply cross-validation.

# EARLY STOPPING AND $L_2$

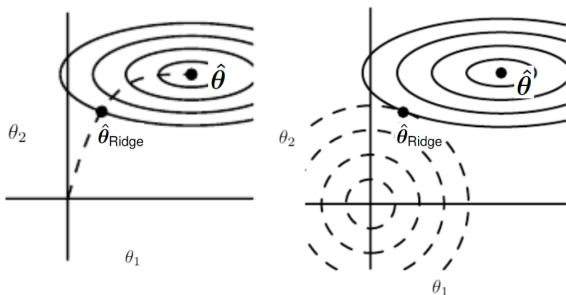
| Strengths                                         | Weaknesses                                                                                       |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Effective and simple                              | Periodical evaluation of validation error                                                        |
| Applicable to almost any model without adjustment | Temporary copy of $\theta$ (we have to save the whole model each time validation error improves) |
| Combinable with other regularization methods      | Less data for training $\rightarrow$ include $\mathcal{D}_{\text{val}}$ afterwards               |

- For simple case of LM with squared loss and GD optim initialized at  $\theta = 0$ : Early stopping has exact correspondence with  $L_2$  regularization/WD: optimal early-stopping iter  $T_{\text{stop}}$  inversely proportional to  $\lambda$  scaled by step-size  $\alpha$

$$T_{\text{stop}} \approx \frac{1}{\alpha\lambda} \Leftrightarrow \lambda \approx \frac{1}{T_{\text{stop}}\alpha}$$

- Small  $\lambda$  ( regu.  $\downarrow$ )  $\Rightarrow$  large  $T_{\text{stop}}$  (complexity  $\uparrow$ ) and vice versa

# EARLY STOPPING AND $L2 / 2$

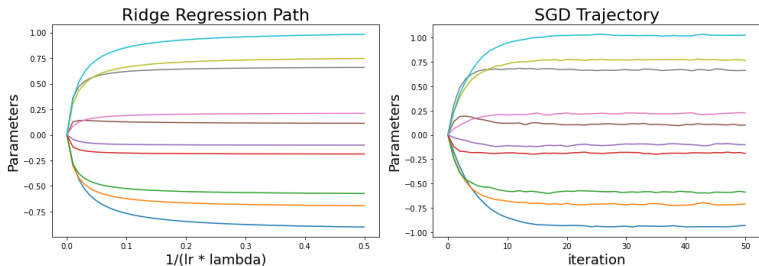


Goodfellow et al., 2016, p. 249 ff.

**Figure:** Effect of early stopping. *Left:* The solid lines indicate contours of the square loss objective. Dashed line indicates trajectory taken by GD initialized at origin. Instead of reaching minimizer  $\hat{\theta}$ , ES results in trajectory stopping earlier at  $\hat{\theta}_{\text{ridge}}$ . *Right:* Effect of  $L2$  regularization. Dashed circles indicate contours of  $L2$  constraint which push minimizer of regularized cost closer to origin than minimizer of unregularized cost.

# SGD TRAJECTORY AND $L_2$

Solution paths for  $L_2$  regularized linear model closely matches SGD trajectory of unregularized LM initialized at  $\theta = 0$



Ali et al., 2020

**Caveat:** Initialization at the origin is crucial for this equivalence to hold, which is almost never used in practice in ML/DL applications